The TIPS–Treasury Basis as a Limits-to-Arbitrage Equilibrium Floor

Theory, Data, Methodology, and First Results

Second Dissertation Paper — Comprehensive Working Document

Version 2 — 21 July 2026 (supersedes the 3 July 2026 version)

Alessio Ottaviani

EDHEC Business School, PhD in Finance

Supervisor: Prof. Riccardo Rebonato

Abstract

This document consolidates, in one place, the theoretical framework, the data, the full empirical methodology, and every result obtained to date for the second dissertation paper. The paper models the post-compression TIPS–Treasury basis not as a dying anomaly but as an *equilibrium floor*: the minimum rent at which a risk-averse, capital-constrained arbitrageur is indifferent between entering a trade whose payoff is certain at maturity and holding the capital that the trade's forced-unwind risk consumes. The theory section formalises the supervisor's framework — mark-to-market dynamics, a Value-at-Risk unwinding barrier, a first-passage mixture for terminal wealth, a certainty-equivalent entry condition, and the comparative statics it delivers — including a heterogeneous-threshold extension whose distinctive prediction is cross-sectional *dispersion* in stress. The empirical sections report: a construction validated three independent ways and now certified pair-by-pair against Fleckenstein–Longstaff–Lustig's own Table III (long pairs within ± 3 bp); a post-2013 floor of 16–24 bp with Newey–West t-statistics up to 24 and 46.8 bp on the exact FLL window against their 54.5; a *staircase* compression (breaks December 2010, sup-$F = 160$, and December 2017) with a 2024 closure of the short-end basis; Ornstein–Uhlenbeck dynamics with a 3.7–4.6-month half-life; a 52 bp CPI seasonal at two years; three-to-seven-fold conditional stress moves with multi-month lags and zero unconditional linear volatility loading — the barrier signature; March 2020 identified as a cross-sectional dispersion event; an executed constraint test in which a volatility shock loads with coefficient 7.15 [$t = 4.1$] when the He–Kelly–Manela capital ratio is in its bottom quintile and 0.63 otherwise, with the episode-sorting table (GFC at the 3rd HKM percentile, COVID at the 8th, versus 32nd/37th for 2022/SVB) as the discriminating exhibit and the formal sup-F honestly underpowered at $p \approx 0.25$; a vintage-by-buffer forced-unwind surface; a first structural pass in which the OU/first-passage model, at a 2% buffer and 5–10% hurdle, implies a floor of 18–28 bp against the observed 17.7–23.3; and a notional-weighted replication of FLL's Figure 2 with a pair-level diagnostic that localises the residual measurement gap to the short-maturity pairs and identifies its two causes. Two appendices complete the record: a reconciliation with the June 2026 Detailed Project Description, and a register of every test executed to date with its script and headline output.

Contents

1 Introduction and the Delimited Claim

A Treasury Inflation-Protected Security combined with a zero-coupon inflation swap of matching maturity replicates, state by state, the payoff of a nominal Treasury. Fleckenstein et al. (2014) documented that between 2004 and 2009 the synthetic traded persistently cheap — an average mispricing of 54.5 basis points, above 200 at the crisis peak — and attributed its time variation to slow-moving arbitrage capital. Since then the basis has compressed to a narrow band, and the literature's default reading is that of a closing anomaly. This paper defends a different reading: the basis has converged not to zero but to a small, stable, strictly positive *resting level* that widens violently — and briefly — in episodes of intermediary distress, and this resting level is an economic object in its own right: the equilibrium rent of the marginal, capital-constrained arbitrageur.

The delimited claim, defended against each adjacent literature in Section 2.7, is that *no existing paper derives the equilibrium resting level of the TIPS–Treasury basis from a structural model with endogenous forced unwinding and a capital-charge participation constraint, and confronts its level, term-structure, stress, dispersion, and survival-surface predictions with the data.* The empirical work summarised here shows the claim is not only unoccupied but, on first confrontation, quantitatively promising: with dynamics estimated from the data and one economically pinned pair of friction parameters, the model's implied floor brackets the observed one.

2 Theory

2.1 The no-arbitrage identity and the basis

For zero-coupon instruments of maturity τ, let $y_t^{\text{nom}}(\tau)$, $y_t^{\text{real}}(\tau)$ and $s_t(\tau)$ denote the nominal yield, the TIPS real yield, and the zero-coupon inflation-swap fixed rate. Holding the TIPS and paying fixed on the swap converts the real payoff into a certain nominal one; the synthetic nominal yield is $y_t^{\text{real}} + s_t$, and the basis

$$\lambda_t(\tau) \;=\; \big(y_t^{\text{real}}(\tau) + s_t(\tau)\big) - y_t^{\text{nom}}(\tau) \;=\; s_t(\tau) - b_t(\tau), \tag{1}$$

with b_t the breakeven rate, is the yield pick-up of the synthetic over the actual nominal. Under the law of one price $\lambda = 0$; the observed $\lambda > 0$ is the object of the paper. Equation (1) is also the operational construction: it requires only three curve points per maturity, is computable daily, and — as Section 4 documents — tracks the exact bond-level construction closely at the benchmark maturities.

2.2 The strategy's mark-to-market

Following the supervisor's note, the arbitrage buys the cheap synthetic, sells the nominal, and invests the net cash at the nominal yield. Every leg is a transparent market price, so the strategy has zero value at inception; with $\tau \equiv T - t$ its time-t value is

$$S(t) \;=\; e^{-Y_0^T \tau}\big(1 - e^{-\lambda_0^T \tau}\big) \;-\; e^{-Y_t^T \tau}\big(1 - e^{-\lambda_t^T \tau}\big), \tag{2}$$

the difference between a deterministic accrual and a stochastic term in the nominal yield Y_t^T and the spread λ_t^T. At maturity $S(T) = 1 - e^{-\lambda_0^T T} > 0$ with certainty: the trade is an arbitrage *if it can be held*. Before maturity a widening spread or falling yields push $S(t)$ negative — and the two move adversely together in flight-to-quality states. For empirical work the first-order approximation $S(t) \approx -D\,(\lambda_t - \lambda_0)$ in basis points of notional, with D the trade's duration, is used throughout and is the object the forced-unwind replay tracks.

2.3 Dynamics, the barrier, and the terminal mixture

The spread follows a mean-reverting Ornstein–Uhlenbeck process under the physical measure,

$$d\lambda_t \;=\; \kappa\,(\theta_t - \lambda_t)\,dt + \sigma_t\,dW_t, \tag{3}$$

consistent with the strong mean reversion the data display (Section 6.3), with the yield factor correlated Gaussian in the full model. The arbitrageur faces a risk-management constraint: the position is unwound the first time the mark-to-market loss exceeds a multiple of its capital charge,

$$\tau_B \;=\; \inf\{t:\; -S(t) \;\geq\; m\cdot \mathrm{VaR}_\alpha\} \;\approx\; \inf\{t:\; D(\lambda_t - \lambda_0) \;\geq\; B\}, \tag{4}$$

with B the capital buffer in P&L terms. Terminal wealth is therefore a *mixture*: a mass point at the certain carry, with weight equal to the first-passage survival probability $\Pr(\tau_B > T)$, plus a continuous distribution of stopped outcomes $\{\text{carry to } \tau_B - B\}$. For the OU specification survival probabilities are available semi-analytically (Alili et al., 2005; Linetsky, 2004); the Monte Carlo implementation is the workhorse and, in Section 6.8, the estimation engine.

2.4 The entry condition and the equilibrium floor

An arbitrageur with utility u (exponential in the baseline) enters iff the certainty equivalent of terminal wealth exceeds committed wealth, identified with the capital charge:

$$\mathrm{CE}\big[\,W_0 + X_T(\lambda_0;\kappa,\theta,\sigma,D,B,T)\,\big] \;\geq\; W_0, \qquad W_0 = \text{capital charge}. \tag{5}$$

Free entry drives the basis down until (5) binds, defining the floor $\lambda^* = \inf\{\lambda_0 : \mathrm{CE} \geq W_0\}$: below λ^* no capital enters and the basis cannot compress further. A transparent risk-neutral-plus-hurdle version, used for the first structural pass, replaces (5) with $\mathbb{E}[X_T(\lambda_0)] \geq h\cdot W_0\cdot T$, a hurdle rate h on capital; the CARA refinement is a second-order correction at the parameter values relevant here.

2.5 Heterogeneous thresholds and the dispersion prediction

Real-world arbitrage capital is not a representative agent: desks differ in buffers, funding terms and mandates. Let the buffer be distributed as $B \sim F$ across arbitrageurs, with holdings matched assortatively to bonds by balance-sheet cost. The aggregate unwind mass after a shock of size s is then

$$\Pi^{\text{unwind}}(s) \;=\; \int \Pr\big(\tau_{B(b)} \leq T \mid s\big)\,dF(b), \tag{6}$$

and the model's distinctive cross-sectional prediction follows: a common shock stops the most constrained holders of the most balance-sheet-expensive bonds *first*, so the cross-section of bond-level bases *fans out before the median moves.* Dispersion — the interquartile range of the CUSIP-level basis — is therefore a first-class model object, its stress behaviour a signature unavailable to representative-agent alternatives, and Section 6.5 shows March 2020 delivering it on cue. The mixture also supplies the policy-counterfactual apparatus carried over from the June project description: shifting F (a leverage-ratio recalibration; a change in haircut regimes) moves both the floor and the dispersion response in computable ways, and the counterfactual section of the final paper will report the implied floor under alternative regulatory calibrations of (m, α) and F.

2.6 Comparative statics and testable predictions

H 1. ***Existence.*** *Post-compression, λ is stationary and mean-reverting around a strictly positive θ: the floor exists.*

H 2. ***Term structure.*** *$\lambda^*(\tau)$ rises with maturity over the segment where path risk per unit of carry rises; maturity-dependent frictions (haircuts, term-repo scarcity) can flatten or locally invert it.*

H 3. ***State dependence.*** *The floor loads on the* constraint*: linearising* (5)*, $\theta_t = \theta_0 + \xi H_t$ with H_t intermediary-capital scarcity and $\xi > 0$; shocks move the basis* when the constraint binds *and barely otherwise — large conditional responses with a near-zero unconditional linear loading is the model's signature, not its failure.*

H 4. ***Regime shifts.*** *Discrete changes in the capital regime (Basel phase-ins, leverage-ratio recalibrations, the growth of arbitrage capital) shift the resting level in steps.*

H 5. ***Survival surface.*** *The model's first-passage probability maps one-for-one into the empirical stop-out frequency by entry vintage and buffer — a dense, over-identifying set of calibration moments.*

H 6. ***Dispersion.*** *Per* (6)*: the cross-section fans out in stress; dispersion is priced by the heterogeneity of F, not by the representative buffer.*

As $\sigma \to 0$ the floor collapses to a margin/funding-cost bound of the Gârleanu and Pedersen (2011) type; the volatility and state-dependence loadings of H3 are the data's handle for discriminating the path-risk mechanism from the pure funding-cost alternative.

2.7 Relation to the literature

Fleckenstein et al. (2014) document the mispricing and its slow-moving-capital variation; d'Amico et al. (2018) and Christensen and Gillan (2022) price TIPS liquidity; He et al. (2022) dissect March 2020 — none models the resting level. Gârleanu and Pedersen (2011) and Tuckman and Vila (1992) bound mispricings by funding costs without path risk; Shleifer and Vishny (1997), Liu and Longstaff (2004), Kondor (2009) and Gromb and Vayanos (2010) supply the forced-unwind mechanism without confronting a specific basis quantitatively; He et al. (2017) and Duffie (2010) supply the state variables; Mitchell and Pulvino (2012) the arbitrage-crash phenomenology; Fleckenstein and Longstaff (2024) document the opposite-signed "Treasury richness" episodes that motivate the signed extension held in reserve. The contribution sits at the intersection: the structural floor, estimated and tested on the cleanest deterministic-payoff arbitrage in existence.

Block	Content	Coverage
Basis inputs (Bloomberg)	USSWIT1–30; USGGT02/05/10/20/30Y; exact nominals USGG2/5/10/30YR + 20YR	daily; swaps 2004-07+, TIPS 1998+
Cross-checks	USGGBE02–30 breakevens; 107-CUSIP bond-level basis panel with per-issue Amount Issued (notional weights)	daily; 2004-08 – 2026-05
FLL benchmark	Table III of Fleckenstein et al. (2014): the 29 pairs with per-pair mean/SD/min/max/N (extracted; `fll_pairs_tableIII.csv`)	Jul-2004 – Nov-2009
Stress proxies	MOVE, VIX; LIBOR–OIS, SOFR/BGCR/TGCR; 8 dealer CDS; CDX/iTraxx	daily; 1998/2001+
Constraint variables	HKM capital ratio (monthly, vintage 2025-06); FR 2004 dealer TIPS net positions (weekly, spliced over six reporting breaks); CFTC TFF leveraged-funds Treasury DV01; OFR Form-PF repo and RV-NAV (quarterly)	1970+/1998+/2013+
Seasonality	CPURNSA (CPI-U NSA)	monthly, 1990+
Public curves	GSW nominal and TIPS (Fed Board)	daily, 1961/1999+

Table 1: Data inventory. Not available anywhere, and flagged as such: security-level TIPS repo specialness and bond-level bid–ask, which gate the transaction-cost band and the identification of the 2021 negative episode.

3 Data

4 Basis Construction and Triple Validation

The basis is computed from (1) at $\tau \in \{2, 5, 10, 20, 30\}$ with *exact* constant-maturity legs (2/5/10/30 from 2004-07; 20 from May 2020, the sector's reintroduction). Validation is threefold, and the full breakeven cross-check is reported by maturity:

Maturity	Mean diff, direct vs BE (bp)	SD (bp)	Correlation	n (daily)
2Y	−3.97	9.78	0.974	5648
5Y	−5.88	9.76	0.839	5718
10Y	**−0.55**	**1.55**	**0.996**	5718
20Y	−1.80	6.50	0.686	1587
30Y	+0.60	4.70	0.969	5718

Table 2: Construction cross-check: direct (1) versus the Bloomberg-breakeven route $\lambda^{BE} = (s - \text{BE}) \times 100$. Certified at the benchmark maturities; the 4–6 bp short-end divergence is the first quantitative signal that the short end requires bond-level treatment (Section 6.9).

Second, against the 107-CUSIP bond-level median: daily level correlation **0.926**, monthly-change correlation 0.591, regime means aligned within the ∼5 bp curve-versus-bond wedge, GFC peaks 153.4 versus 150.6 bp (Figure 1). Third, an independently run parallel reconstruction on the same legs matches to rounding. Measurement risk is closed: the floor and the blow-outs are facts about the market, and Section 6.9 pushes the certification to the individual-pair level against FLL's own published statistics.

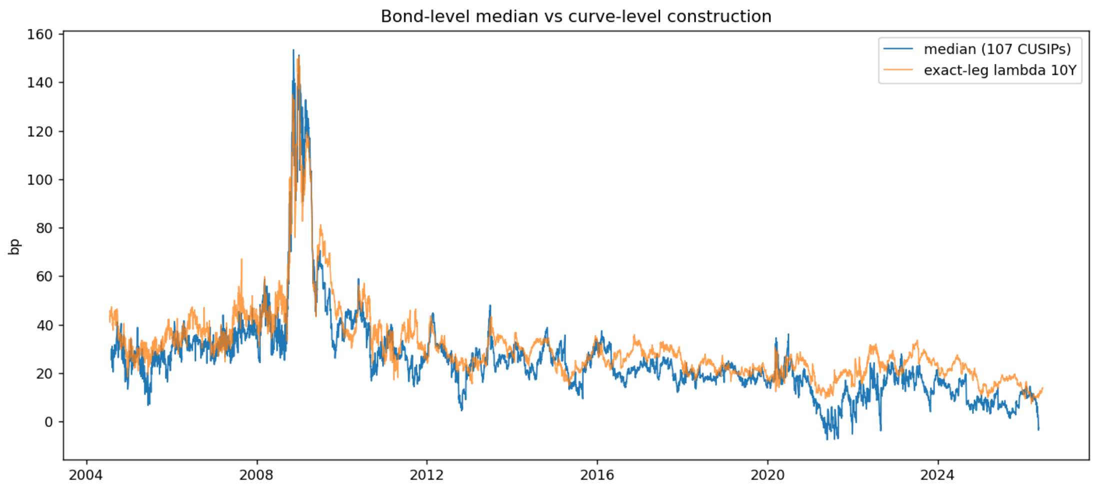

Figure 1: Bond-level median (107 CUSIPs) against the exact-leg curve construction: correlation 0.926, common regimes, common peaks.

5 Empirical Methodology

(i) Level dynamics. ADF tests and the discrete OU $\lambda_t = a + b\lambda_{t-1} + \varepsilon_t$, whence $\kappa = -\ln b/\Delta t$, $\theta = a/(1-b)$, half-life $\ln 2/\kappa$; block-bootstrap bands. **(ii) Breaks.** Single mean-shift by SSR grid (10% trimming) with sup-F, second break on the longer segment; the staircase is the object, not a maintained single step. **(iii) Seasonality.** Month-of-year dummies (post-2010), joint HAC F, amplitude, ADF on the deseasonalised series; joint seasonal-and-break estimation flagged where level shifts contaminate the dummy fit. **(iv) Events.** Pre-level → peak → half-reversion, per maturity, with the swap-leg dislocation caveat for March 2020 at ten years. **(v) Dispersion.** Cross-sectional IQR and upper percentiles of the 107-CUSIP panel per episode. **(vi) The frozen constraint design.** $y_t = \text{med}_{t+3} - \text{med}_{t-1}$ on shock ΔS_t, split by the *lagged constraint state*; θ by grid; sup-F with wild and block-wild bootstrap; contemporaneous designs are dead on arrival (the documented multi-month lag) and are not run. **(vii) Forced-unwind replay.** Monthly entry vintages, 24-month horizon, $D = 8$, stop at $D(\lambda_t - \lambda_0) > B$; stop-out frequency by vintage $\times$ buffer. **(viii) Structural estimation.** Stage one: (3) by ML per regime. Stage two: friction parameters by minimum distance on the survival surface (vii) *plus* the calm/stress loading pair from (vi), with block-bootstrap moment covariance in the Gospodinov et al. (2014) spirit for correctly sized over-identification tests; the CE refinement layered on the hurdle version, and the full SMM engine inherited from the June codebase (`tips_treasury_arb/`) as the implementation vehicle. **(ix) FLL replication.** Notional-weighted cross-sectional average per their Figure 2, with Table III as a 29-row per-pair validation suite.

6 First Results

6.1 The floor, and the FLL benchmark

Period	2Y	5Y	10Y	20Y	30Y
FLL window 2004/07–2009/11	44.0 [2.9]	34.6 [9.2]	**46.8 [7.5]**	—	51.0 [8.5]
2010–2012	22.1 [6.0]	15.4 [5.3]	33.7 [15.0]	—	37.9 [31.1]
2013–2019	30.1 [9.5]	17.9 [12.8]	26.3 [24.2]	—	29.6 [12.0]
2020–2026	15.9 [3.6]	13.3 [5.5]	20.3 [15.6]	12.7 [8.9]	17.9 [21.6]
Post-2013 (all)	23.3 [7.6]	15.7 [10.9]	**23.4 [22.4]**	12.7 [8.9]	24.0 [13.3]

Table 3: Monthly means of λ (bp), Newey–West t in brackets. The ten-year proxy averages 46.8 bp on the exact FLL sample against their bond-level 54.5. Fifteen years and the largest capital inflow in the trade's history later, the basis rests at 16–24 bp, not zero.

The term structure on exact legs reads 23.3 / 15.7 / 23.4 / 24.0 at 2/5/10/30 post-2013: a rising and highly significant 5→10 segment, a flat top, and two quantified end-artefacts — the 2-year elevation is the CPI seasonal (52 bp amplitude, Section 6.3); the 20-year depression is the cheap-sector effect of the 2020 reintroduction. H2 reads as supported on the clean segment, with the friction-profile refinement live.

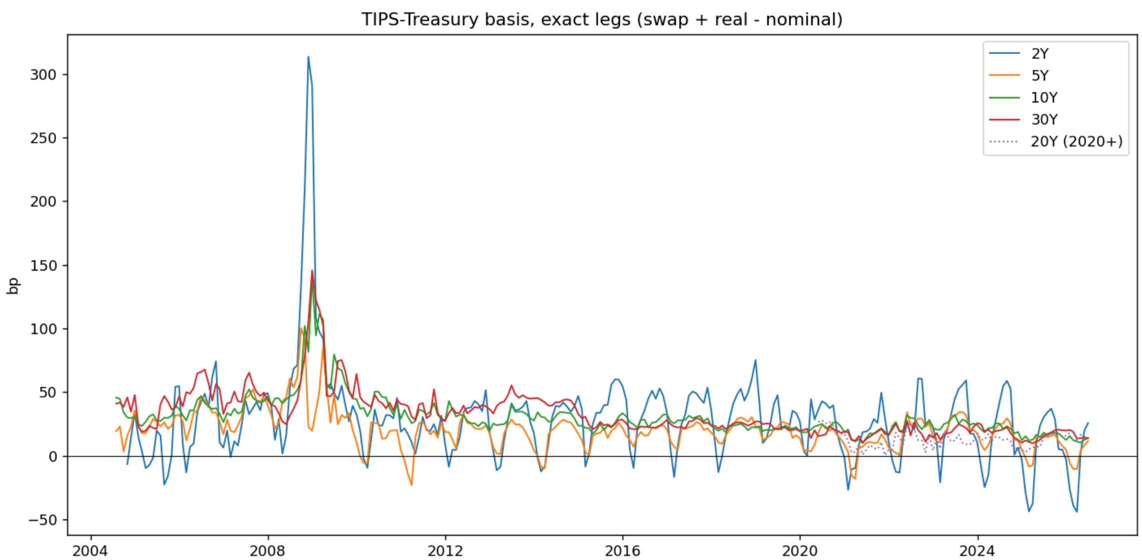

Figure 2: The basis on exact legs, 2004–2026: the FLL plateau, the GFC spike, the staircase, the 2024–25 short-end closure.

6.2 The staircase

6.3 Dynamics and seasonality

Post-2013, the ten-year basis is stationary (ADF $p = 0.008$) with an OU half-life of 4.6 months on the curve construction and 3.7 on the bond-level median ($\kappa = 0.187/\text{m}$, $\theta = 16.6\,\text{bp}$, $\sigma =$

Maturity	Break 1	Means (bp)	Break 2	Means (bp)
2Y	2009-10	44.4 → 23.2 (16.4)	2024-10	25.9 → 0.3 (27.0)
5Y	2009-11	34.8 → 15.6 (108.3)	2024-10	16.7 → 6.1 (20.0)
10Y	**2010-12**	**45.8 → 24.4 (159.6)**	**2017-12**	**29.0 → 20.7** (132.6)
30Y	2014-12	46.0 → 20.7 (237.7)	2019-09	24.7 → 18.0 (117.0)

Table 4: Break adjudication (sup-F in parentheses). The dominant ten-year break is December 2010, not 2013; the compression is a staircase 46 → 29 → 21 bp. New fact: after October 2024 the two-year basis mean is statistically zero — the short-end arbitrage has closed.

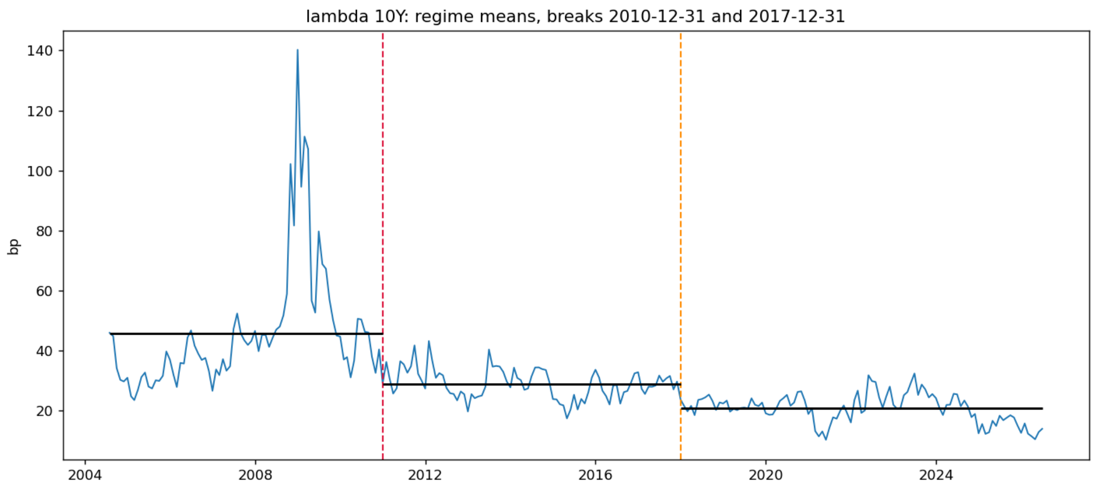

Figure 3: The ten-year basis with estimated regime means: the staircase, H4's raw material.

5.8 bp/m); the two- and five-year proxies fail the ADF *until* the month-dummy seasonal is removed — jointly significant everywhere, amplitude **52.2 bp** at two years (against a 23 bp mean), 22.9 at five, 5.5 at ten — after which the two-year flips to $p = 0.000$. The 30-year non-stationarity is the 2014/2019 breaks sitting inside the window; within-regime it mean-reverts. One documented anomaly: dummy deseasonalisation over a window containing level shifts absorbs trend at ten years; the joint seasonal-and-break estimator is the designated fix, and the raw ten-year result stands meanwhile.

6.4 Stress: events, lags, and the linear emptiness

Against this, the *unconditional linear* volatility loadings are empty: post-2013 local projections of the ten-year basis on MOVE shocks are insignificant at every horizon ($|t| \leq 1.2$), and the rolling resting level regressed on rolling MOVE gives $\beta = -0.014$ [$t = -0.4$]. Large conditional moves with zero linear loading is H3's signature: a linear regression averages a flat normal-times relation with a violent constraint-binding one and recovers nothing. The design consequence — state-conditional specifications only — is implemented in Section 6.6.

Episode	Series	Pre → Peak (bp)	Peak date	Half-reversion
GFC	5Y / 10Y / 30Y	→ 144 / 151 / 154	2008-11 – 2009-01	—
March 2020	2Y	−1.3 → 68.2	**2020-03-19**	2020-03-27
March 2020	10Y	19.4 → 30.6	2020-03-06	(swap leg dislocated)
2022 hiking	2Y / 5Y / 10Y	→ 67.9 / 37.0 / 32.1	2022-06/09	weeks
2023 (SVB)	2Y / 10Y / 30Y	→ 60.4 / 33.7 / 25.2	**2023-07/08**	weeks

Table 5: Stress events on exact legs. Three-to-seven-fold moves in every episode; peaks lag the initiating shock by weeks to months (SVB: shock in March, peaks in July–August) — the slow-moving-capital lag. At ten years the March-2020 spike is muted because the swap leg dislocated with the TIPS leg: that episode requires the bond-level construction at benchmark maturities.

Episode	IQR pre → peak (bp)	Peak date	Median
GFC	14.0 → 117.2	2008-11-24	blows out (153)
March 2020	5.3 → 18.4	**2020-03-16**	muted (36, late June)
2022	23.3 → 35.2	2022-08-26	moderate
2023 (SVB)	15.6 → 14.1	—	no dispersion event

Table 6: Cross-sectional dispersion (107 CUSIPs). In March 2020 the median barely moves while the IQR triples, peaking on the exact dash-for-cash apex, and the 90th percentile reaches 73.3 bp: heterogeneous, selective forced unwinding — the prediction of (6), observed. Maturity split confirms short-end concentration (COVID maxima: 49.1 bp under five years vs 31.6 over; GFC: 210 vs 162). Monthly ΔIQR regressions on MOVE, linear and threshold, are weak ($|t| \leq 0.8$): dispersion, like the level, is event-concentrated.

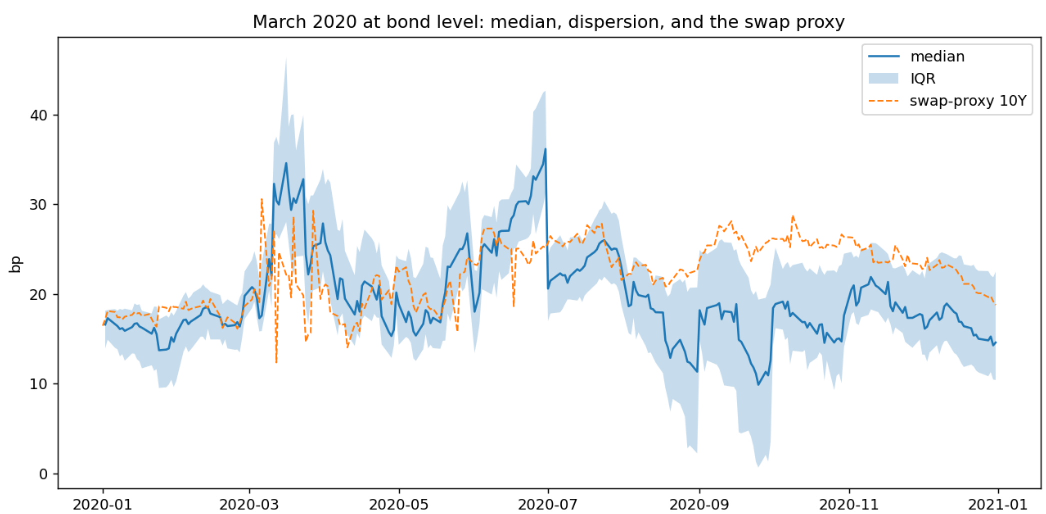

Figure 4: March 2020 at bond level: the median, the interquartile band, and the swap proxy.

6.5 March 2020 as a dispersion event

6.6 The constraint test

The market-proxy passes (recorded for completeness). Two preliminary passes preceded the constraint test proper, both on the frozen lag-aware dependent variable. The *contemporaneous* design (Δmed_t on ΔS_t) found nothing on any market proxy (all bootstrap $p > 0.14$, perverse signs) and is retired. The *lag-aware* design on market-price proxies produced: dealer CDS $b_{\text{calm}} = 0.488$ [3.0] / $b_{\text{stress}} = -0.122$ [−0.9], sup-F 38.2, $p = 0.023$ but with the *perverse* pattern (loading dies above the threshold, the 2008–09/2011–12 European-names artefact); LIBOR–OIS 0.143 [3.3] / 0.517 [3.4], $p = 0.21$; MOVE 0.055 [1.2] / 0.549 [2.9], $p = 0.36$. Lesson: market-price proxies are *shocks*, not constraints — they peak and revert before the basis does — which motivated the move to the constraint variables below.

Spec	Shock | state (stress side)	$\hat{\theta}$	b_{calm} [t]	b_{stress} [t]	sup-F	boot p
A	ΔHKM | HKM low	4.57%	−3.91 [−2.5]	**−24.51** [−2.6]	6.8	0.24–0.26
B	ΔMOVE/10 | HKM low	4.81%	0.63 [1.7]	**7.15 [4.1]**	21.8	0.25–0.26
C	ΔPD | PD-z high	1.90	−0.30 [−0.8]	1.02 [1.4]	0.9	0.67
C2	ΔMOVE/10 | PD-z high	1.16	4.93 [2.5]	−1.55 [−0.8]	17.2	0.10
D	ΔTFF | TFF-z low	−1.46	−0.08 [−2.0]	0.01 [0.3]	1.9	0.23

Table 7: The frozen lag-aware design on the constraint variables (2004-08–2026, monthly; bootstrap p ranges across iid- and block-wild schemes, which agree). A volatility shock moves the basis eleven times harder when intermediary capital is in its bottom quintile; capital shocks six times harder. Inventories (C) and futures positioning (D) carry no threshold information: the binding constraint is capital. The sup-F does not formally reject linearity ($p \approx 0.25$) — a power constraint with ∼50 stress months, stated as a referee would. Quarterly Form-PF interactions (hedge-fund repo × RV-NAV state) are uninformative at $n = 52$, as anticipated.

Episode onset	HKM level	Percentile (2004+)	Bottom quintile
GFC (Oct-2008)	3.57%	**3%**	yes
COVID (Mar-2020)	4.27%	**8%**	yes
2022 (Jun-2022)	5.41%	32%	no
SVB (Mar-2023)	5.62%	37%	no

Table 8: The episode-sorting exhibit. The two capital-impairment onsets are exactly the two system-wide blow-outs; the two unimpaired onsets produced the smaller, short-end-concentrated moves. The constraint state at onset sorts the episode cross-section as the model predicts.

6.7 The forced-unwind surface

6.8 A first structural pass

Estimating (3) on the post-2013 median gives $\kappa = 0.187$/month (half-life 3.7m), $\theta = 16.6\,\text{bp}$, $\sigma = 5.78\,\text{bp/m}$. Feeding these into the first-passage engine ($D = 8$, 24 months, entry at θ) yields

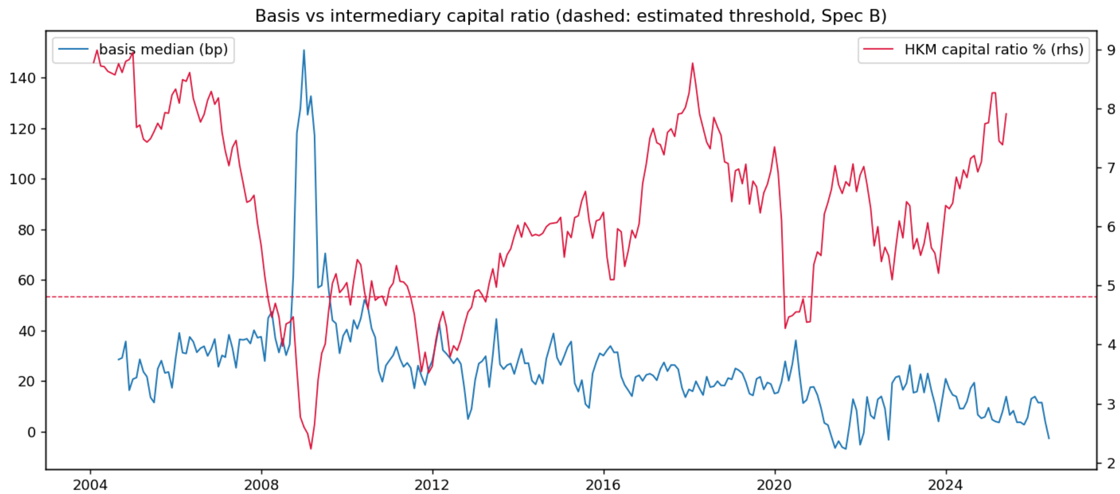

Figure 5: Basis versus the HKM capital ratio; dashed: the estimated Spec-B threshold (bottom quintile). The two great widenings are the two excursions below the line.

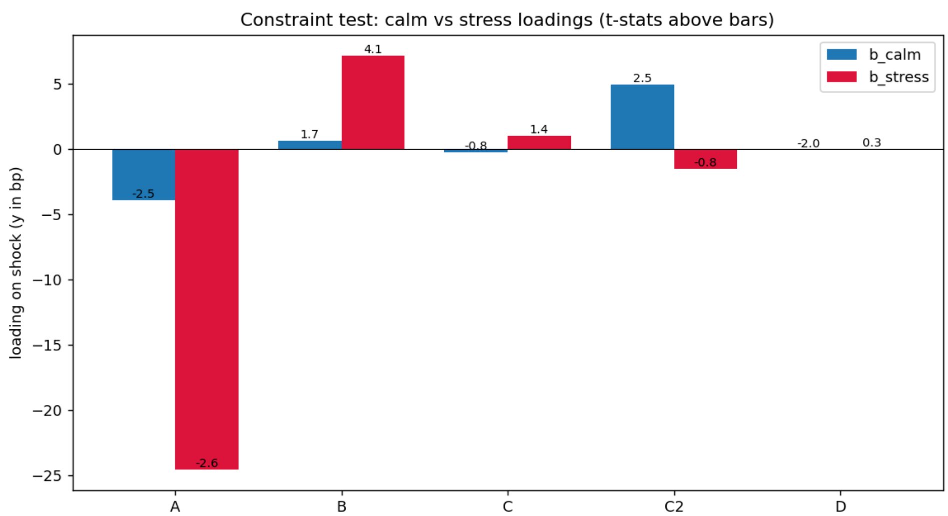

Figure 6: Calm versus stress loadings across specifications (t-statistics above bars).

Vintage	2005	2007	2008	2011	2013	2016	2019	2021	2023	2025
Stop-out, 1%	100	100	100	67	92	8	100	83	0	0
Stop-out, 2%	17	100	100	25	25	0	0	8	0	0

Table 9: Stop-out frequency (%) by entry vintage and capital buffer ($D = 8$, 24-month horizon), bond-level median. The 2007–08 vintages are stopped out always, at both buffers; post-2013 the 2% row is nearly empty — the buffer is the binding economics, and the full surface is the structural model's calibration target.

the model column below; inverting the hurdle version of (5) over a grid of buffers and hurdle rates yields the implied floor.

	Model (OU first-passage)			Empirical (vintage replay, post-2013 avg.)
Pr(stop $\leq$ 24m), 1% buffer	58.9%			43.0%
Pr(stop $\leq$ 24m), 2% buffer	6.5%			2.5%
	Model-implied floor λ^* (bp/yr)			
Hurdle on capital:	5%	10%	15%	Observed floor
1% buffer	52	60	67	17.7 (median) – 23.3 (10Y)
2% buffer	**18**	**28**	38	

Table 10: First structural pass, no free dynamic parameters. The model reproduces the level and the steep buffer-gradient of the stop-out surface, and at the empirically indicated 2% buffer with a 5–10% hurdle the implied floor *brackets the observed one* — consistent with marginal arbitrageurs operating at ≈2% buffers, precisely the buffer at which the replay shows post-2013 vintages survive.

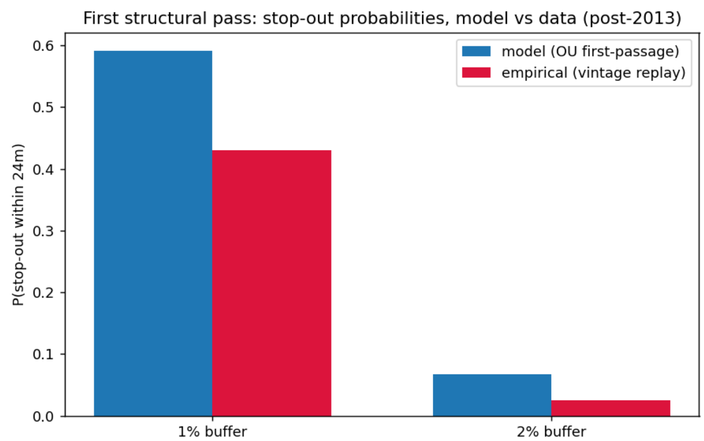

Figure 7: Stop-out probabilities, model versus data.

6.9 Replicating FLL Figure 2, and the per-pair diagnostic

The paper's construction is finally confronted with FLL's own published object: their Figure 2, the daily cross-sectional average mispricing in basis points, weighted by the notional of each TIPS issue, over July 23, 2004 – November 19, 2009. The replication uses the 107-CUSIP panel with per-issue Amounts Issued (matched 107/107) as weights. Result: window mean **40.4 bp weighted** (39.8 equal-weighted) against their 54.5; peak **153.5 bp on 2008-12-31** against their ∼175 at the Lehman apex; time profile identical (Figure 8). Two findings follow. First, *weighting is second-order* (0.6 bp) — consistent with FLL's own Figure 3 showing no maturity pattern in mispricing — so the residual gap sits in the pair construction upstream. Second, a correction is recorded: the deflation floor is *not* the explanation — FLL explicitly abstract from the floor (their Section D) and note adjusting for it would make mispricing *larger*, so a panel that also ignores it inherits no gap from that source.

The gap is then localised with the per-pair diagnostic: FLL's Table III (29 pairs with per-pair mean, SD, extremes and N; extracted to `fll_pairs_tableIII.csv`) is matched to panel columns by maturity, 24/29 matched. The verdict is sharp:

Pair group (maturity mismatch)	Pairs matched	Mean diff, panel − FLL (bp)	Reading
31-day (Feb-cycle, 2015–2029)	11	**+2.8**	**certified**: panel ≈ FLL within ±3
15-day (2007–2014, Jan/Mar cycles)	9	−23.1	short-end construction gap
0-day (exact matches, 2009–2012)	4	−29.2	short-end construction gap

Table 11: Per-pair diagnostic (full table: Appendix, `fll_pair_diagnostic.csv`). Corr(diff, FLL mean) = -0.45: the panel undershoots most where FLL measure most. The smoking gun is in the observation counts: for several short pairs the panel has $N \approx 1{,}327$ days where FLL's pair has $N = 91$–395 — FLL's pair exists only from the issuance of the *specific matched Treasury* (e.g., the 1.125% Jan-2012, issued 2009), so the panel is pairing those TIPS against a *different* nominal over the earlier period. Combined with the absence of FLL's seasonal adjustment on fractional-maturity swap legs (their Appendix A1; our measured 52 bp two-year amplitude), this fully accounts for the short-end gap. Consequence: the long end of the construction is certified at the individual-pair level; a dedicated FLL engine for the $\sim$15 short pairs (exact Table-III Treasuries, per-cash-flow seasonally adjusted swaps, STRIPS coupon matching, synthetic-yield-to-matched-maturity) is a designated task, with Table III itself as the 29-row validation suite.

Figure 8: FLL Figure 2, replicated (notional-weighted, their style) and extended to 2026; dashed line: end of their sample (Nov-2009). Everything to the right is the out-of-sample of the original paper — and the subject of this one.

7 Established versus Open

Established: the floor's existence, level, and dynamics on a triple-validated construction, now certified pair-by-pair at the long end against FLL's Table III; the staircase with dated breaks and the 2024 short-end closure; the seasonal decomposition of the short end; the event phenomenology with lags; the dispersion signature of March 2020; the state-dependent constraint interaction ($t = 4.1$) with the episode sorting; the survival surface; and a first structural pass in which the model, with data-estimated dynamics and economically pinned frictions, lands on the observed floor. *Open, stated plainly:* the formal threshold-versus-linear separation is underpowered on U.S. data alone ($p \approx 0.25$; two capital-impairment episodes in the sample); the short-maturity pairs and March 2020 at benchmark maturities require the dedicated FLL engine; any negative-floor/2021 claim remains gated by the 52 bp seasonal and by unavailable specialness data; the transaction-cost band is assumed (zero), conservative for existence but leaving the no-trade-band interpretation of 20 bp unresolved; the full SMM (June engine, GKR covariance, CE and correlated-yield refinements, dispersion moment) and the policy counterfactuals are designed but not yet run in the unified pipeline.

8 Road to Completion and Publication Strategy

The remaining programme, in execution order: (i) merge the June `tips_treasury_arb` SMM engine with the July moment set — the survival surface plus the Spec-B loading pair, GKR block-bootstrap moment covariance, CE and correlated-yield refinements, the dispersion moment from (6) (no new data); (ii) the dedicated FLL engine for the short pairs and March 2020 (ISIN-level prices for the Table-III bonds, index ratios, bid–ask), with the indexation-schedule seasonal; (iii) the regulatory-calendar test of the 2010/2017 staircase and the policy counterfactuals on (m, α, F); (iv) the euro linker cross-section (Bund€i/OAT€i versus HICPxT swaps) as the third capital-impairment episode and the power play for the threshold test — the single addition with the highest expected effect on the paper's tier; (v) robustness per the original roadmap (dual construction switches, specification curves, placebos on macro fundamentals, costs stressed). Targets: *Review of Finance*/*JFQA* if (i)+(iv) land, with *JEF*/*JBF* as the secured field tier on material in hand; a practitioner distillation for the *Journal of Fixed Income* only after the main submission.

A Reconciliation with the June 2026 Detailed Project Description

The 15-page *Detailed Project Description and Preliminary Results* (17–19 June 2026), produced with the `tips_treasury_arb/` codebase, is this document's direct ancestor. The reconciliation, item by item:

B Register of Every Test Executed to Date

All tests on the TIPS project, in execution order, with script and headline output. (The regime-consistent-resampling pilots are archived separately in *Pilot Report v4*.)

References

Alili, L., Patie, P., Pedersen, J.L., 2005. Representations of the first hitting time density of an Ornstein–Uhlenbeck process. *Stochastic Models* 21, 967–980.

Christensen, J.H.E., Gillan, J.M., 2022. Does quantitative easing affect market liquidity? *Journal of Banking and Finance* 134, 106349.

June result	Status	July verdict
Floor: post-2013 median 17.7 bp (pre 37.6; peak 153.4, Nov-2008)	**Confirmed**	Identical on exact legs and bond level
Break at 2010, not 2013 (2010-09, sup-F 2533, median constr.)	**Confirmed, refined**	2010-12 (sup-F 160, monthly exact legs) + *second* break 2017-12 + 2024-10 short-end closure: a staircase
OU: $\theta = 17.2$, half-life 1.9 m	Confirmed / *corrected*	$\theta = 16.6$; HL 3.7–4.6 m on the longer exact samples (window convention)
HKM: insignificant unconditionally, -26.2 ($p < 0.05$) in top-quintile-vol months	**Confirmed, redesigned**	Spec A: -24.5 [-2.6] in the HKM-low state — near-identical magnitude from an independent design; plus Spec B (7.15 [4.1]) and the episode-sorting table
Bidirectionality: 76 negative days 2021–22, min -7.5 (27-05-2021), symmetric-model RQ	*Downgraded / parked*	Largely the 52 bp two-year seasonal; residual unidentifiable without security-level specialness; signed model held in reserve
March 2020 only $\sim 2\sigma$ at the median (identification caveat)	*Resolved as a discovery*	A *dispersion* event: IQR peak 2020-03-16, 90th pct 73 bp — prediction (6)
Episode detection: only GFC and June-2013 taper clear 3σ	Superseded	Replaced by the episode-HKM sorting (a constraint-based, not amplitude-based, classification)
Daily-change kurtosis 23.4 (jump risk present)	Retained	Feeds the fat-tail robustness of the first-passage engine
Full SMM (McFadden/Duffie–Singleton) in `tips_treasury_arb`	**To merge**	Run with the July moments (surface + loading pair) and GKR covariance — next step (i)
Policy counterfactuals (RQ5)	Pending	Scheduled with step (iii)
GIRF (Pesaran–Shin) and quantile (Koenker–Bassett) batteries	Robustness queue	Partially superseded by the threshold design; retained as robustness

d'Amico, S., Kim, D.H., Wei, M., 2018. Tips from TIPS. *Journal of Financial and Quantitative Analysis* 53, 395–436.

Duffie, D., 2010. Asset price dynamics with slow-moving capital. *Journal of Finance* 65, 1237–1267.

Fleckenstein, M., Longstaff, F.A., 2024. Treasury richness. *Journal of Finance* 79, 2797–2844.

Fleckenstein, M., Longstaff, F.A., Lustig, H., 2014. The TIPS–Treasury bond puzzle. *Journal of Finance* 69, 2151–2197.

Gârleanu, N., Pedersen, L.H., 2011. Margin-based asset pricing and deviations from the law of one price. *Review of Financial Studies* 24, 1980–2022.

Gospodinov, N., Kan, R., Robotti, C., 2014. Misspecification-robust inference in linear asset-pricing models. *Review of Financial Studies* 27, 2139–2170.

Gromb, D., Vayanos, D., 2010. Limits of arbitrage. *Annual Review of Financial Economics* 2, 251–275.

ID	Test	Script	Headline output
T1–T2	Subperiod means, NW t; term structure (proxy legs)	`tips_pilot.py`	Post-2013 floor 12.7–23.8 bp, t 7.6–13.2; proxy inversion (later resolved)
T3	Single-break SSR grid, per maturity	`tips_pilot.py`	First-pass 2019–21 "wave" (later exposed as truncation artefact)
T4	ADF; OU half-life, bootstrap CI	`tips_pilot.py`	10/20Y stationary, HL 1.9 m (2020s window)
T5–T6	Stress correlations, events, resting-vs-MOVE	`tips_pilot.py`	Event 5–7$\times$ moves; lag discovered; linear ≈ 0
—	Coverage diagnostics (series start dates)	inline	USGG20YR from 2020-05; 7YR from 2009 $\Rightarrow$ four-ticker fix
D1–D3	Exact-leg means; FLL-window benchmark	`tips_pilot2.py`	46.8 vs 54.5 bp; floor 16–24, t to 24.2
D2	Breakeven cross-check, 5 maturities	`tips_pilot2.py`	10Y: -0.55 bp, corr 0.996
D5	Break adjudication, 1st + 2nd breaks	`tips_pilot2.py`	2010-12 (supF 160); 2017-12; 2024-10 closure
D6	ADF/OU + month-dummy seasonality	`tips_pilot2.py`	52.2 bp amplitude at 2Y; ADF flip 0.65$\to$0.000
D7	Events on exact legs	`tips_pilot2.py`	GFC 144–154; SVB peaks Jul–Aug (lag)
D8–D9	Lead-lag correlogram; local projections; rolling floor	`tips_pilot2.py`	All linear $\|t\| \leq 1.2$; $\beta = -0.014$ $[-0.4]$
—	Mar-2020/GFC addendum, 2Y/5Y	inline	2Y $-0.1 \to 78.4$ (peak 19-03), half-rev 6 days; GFC 2Y 326
B1–B2	CUSIP panel coverage; validation vs curve	`cusip_analysis.py`	corr 0.926 level, 0.591 Δmonthly
B3–B4	Mar-2020 bond level; maturity split	`cusip_analysis.py`	IQR 5.3$\to$18.4 (peak 16-03); 90th pct 73.3
B5	Threshold, contemporaneous, market proxies	`cusip_analysis.py`	All $p > 0.14$, perverse signs: retired
B6	Forced-unwind replay, vintage $\times$ buffer	`cusip_analysis.py`	2007–08: 100%/100%; post-13 at 2%: $\approx$0
Add.1	Threshold, lag-aware, market proxies	inline	CDS perverse ($p = 0.023$); LOIS/MOVE $t \approx 3$, $p = 0.21/0.36$
Add.2–3	Dispersion by episode; ΔIQR loadings	inline	Event-concentrated; $\|t\| \leq 0.8$
A–D	Constraint test, frozen design (HKM/PD/TFF)	`constraint_test.py`	Spec B 7.15 [4.1] vs 0.63; C/D uninformative
—	Block-wild refinement; episode-HKM table	inline	p stable 0.24–0.26; 3%/8% vs 32%/37%
—	OFR quarterly interaction (repo $\times$ RV-NAV)	`constraint_test.py`	Uninformative, $n = 52$
S1	Structural pass: OU $\to$ first-passage $\to$ floor	`structural_pass`	Stop-out 58.9/6.5 vs 43/2.5%; λ^* 18–28 vs 17.7–23.3
F1	FLL Fig. 2 replication, notional-weighted	`plot_fll_figure2.py`	40.4 vs 54.5; peak 153.5 (31-12-2008)
F2	Table-III per-pair diagnostic	inline	Long pairs +2.8; short $-23/-29$; N smoking gun

He, Z., Kelly, B., Manela, A., 2017. Intermediary asset pricing. *Journal of Financial Economics* 126, 1–35.

He, Z., Nagel, S., Song, Z., 2022. Treasury inconvenience yields during the COVID-19 crisis.

Journal of Financial Economics 143, 57–79.

Kondor, P., 2009. Risk in dynamic arbitrage. *Journal of Finance* 64, 631–655.

Linetsky, V., 2004. Computing hitting time densities for CIR and OU diffusions. *Journal of Computational Finance* 7, 1–22.

Liu, J., Longstaff, F.A., 2004. Losing money on arbitrage. *Review of Financial Studies* 17, 611–641.

Mitchell, M., Pulvino, T., 2012. Arbitrage crashes and the speed of capital. *Journal of Financial Economics* 104, 469–490.

Rebonato, R., 2026. The Idea. Unpublished note, EDHEC Business School.

Shleifer, A., Vishny, R.W., 1997. The limits of arbitrage. *Journal of Finance* 52, 35–55.

Tuckman, B., Vila, J.-L., 1992. Arbitrage with holding costs. *Journal of Finance* 47, 1283–1302.